

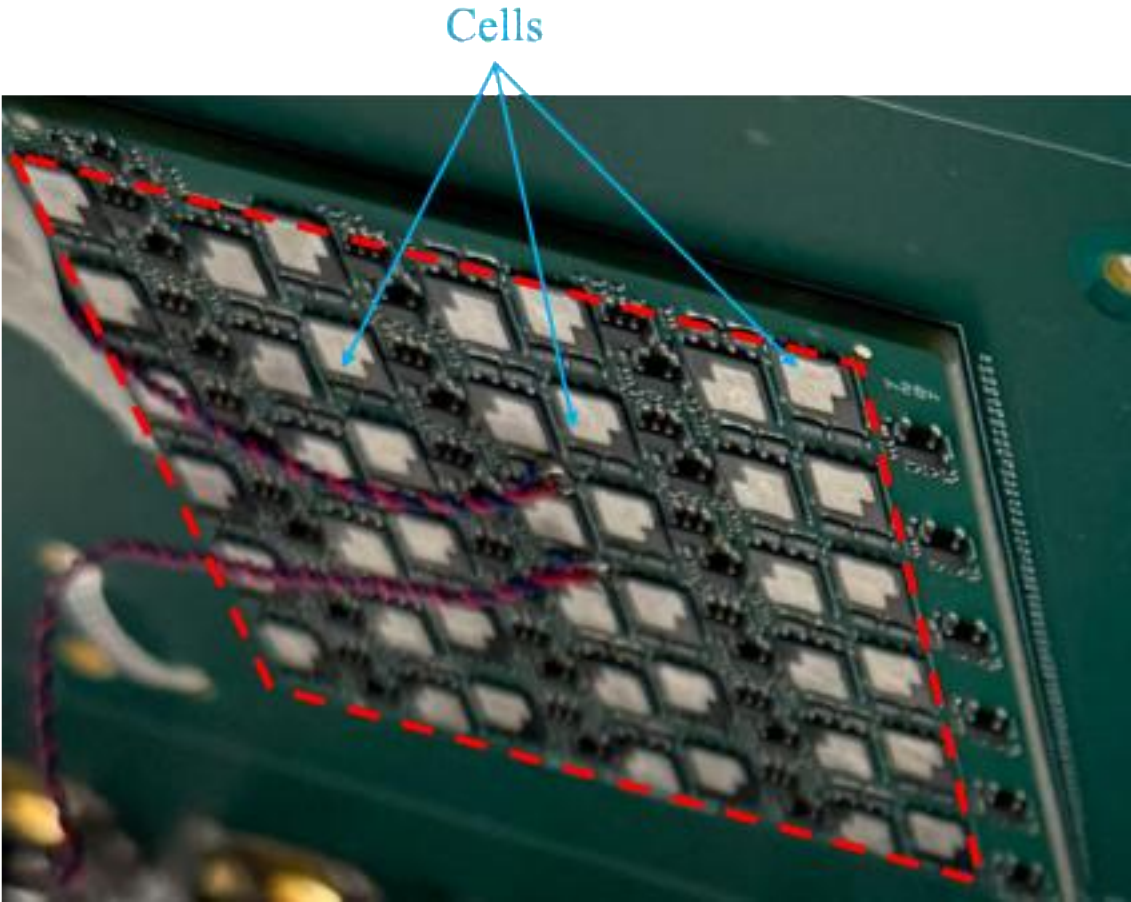
Exhibit 31

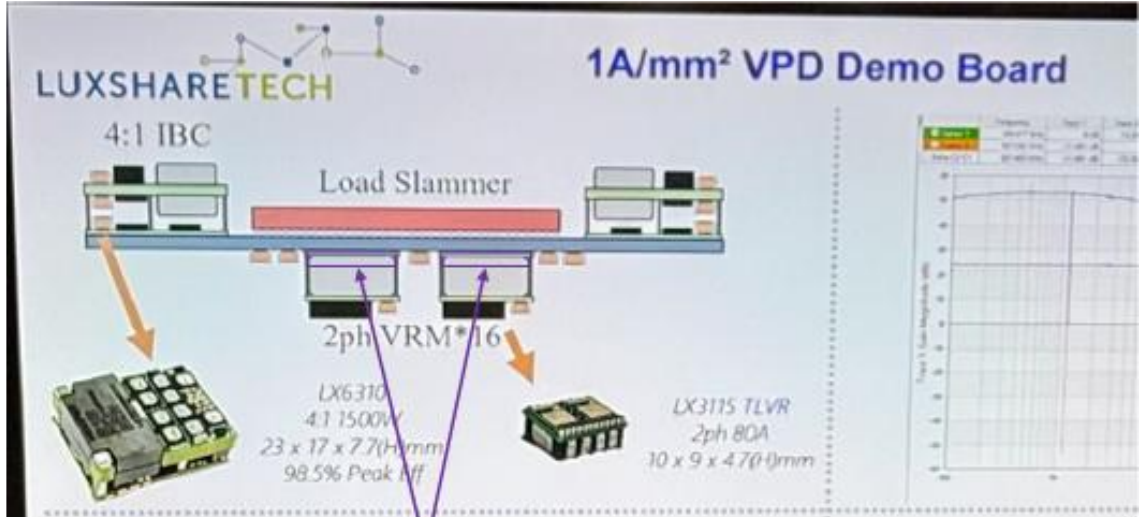
Claim Chart for the Practice of Independent Claims 1, 2, and 40 of Complainant Vicor Corporation's United States Patent No. 10,903,734 by Luxshare VPD Power Module ("Luxshare VPD Module") and Circuit Board Assemblies and Computing Systems Containing the Same¹

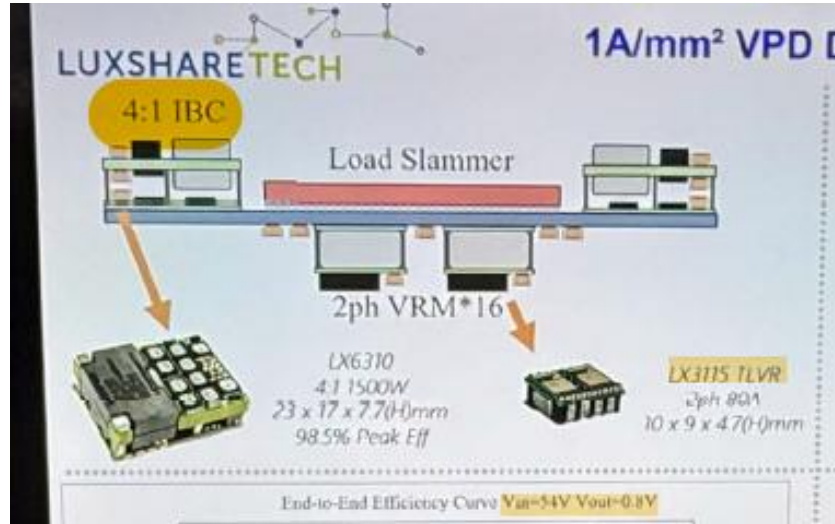
Claim 1

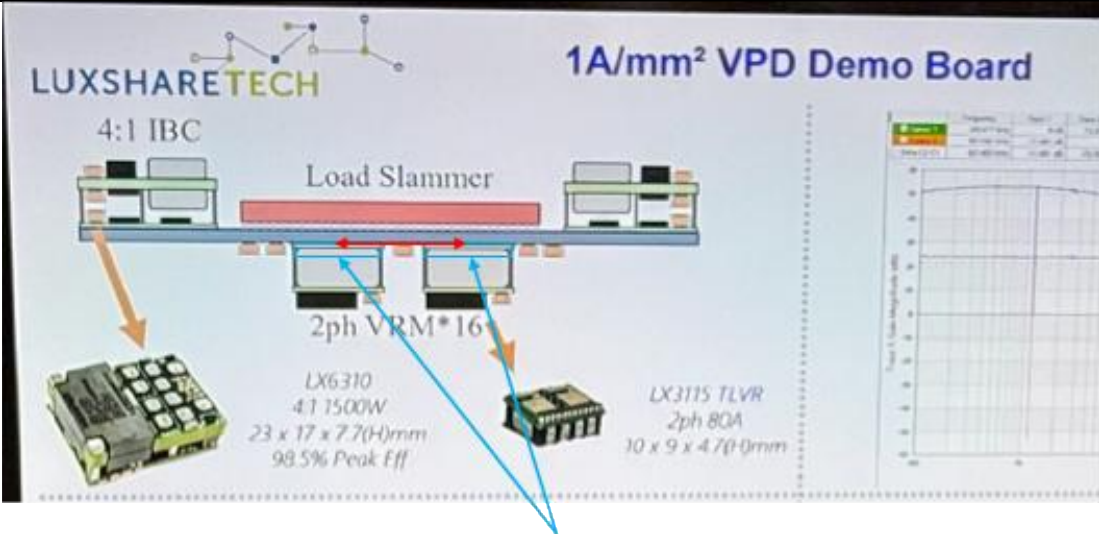
#	Limitation	Practice Evidence
1.pre	A method of powering a semiconductor device having power input terminals, the method comprising:	The Luxshare VPD Module practices "[a] method of powering a semiconductor device having power input terminals." Ex. 32 at 1.

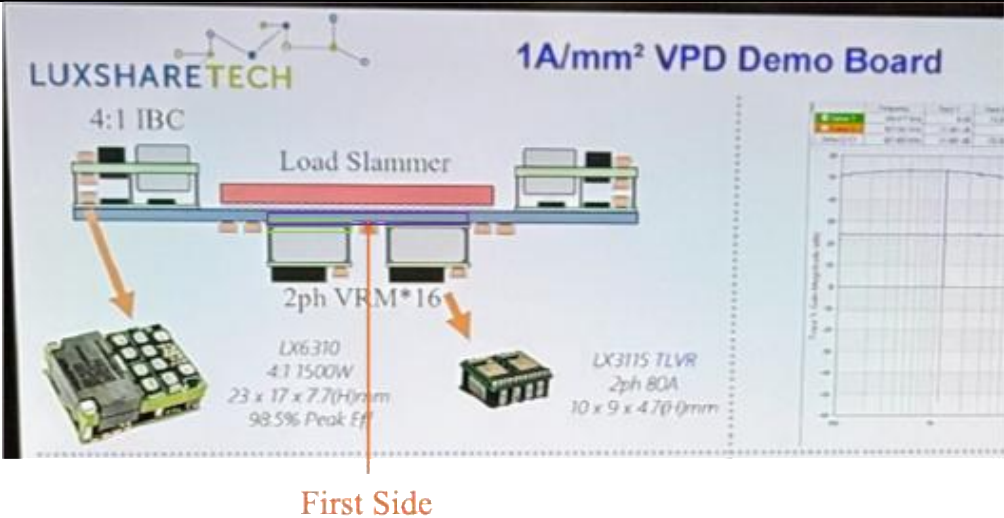
¹ The following analysis of Luxshare VPD Module is applicable to all other equivalent Luxshare power modules as well as to circuit board assemblies and computing systems containing the same.

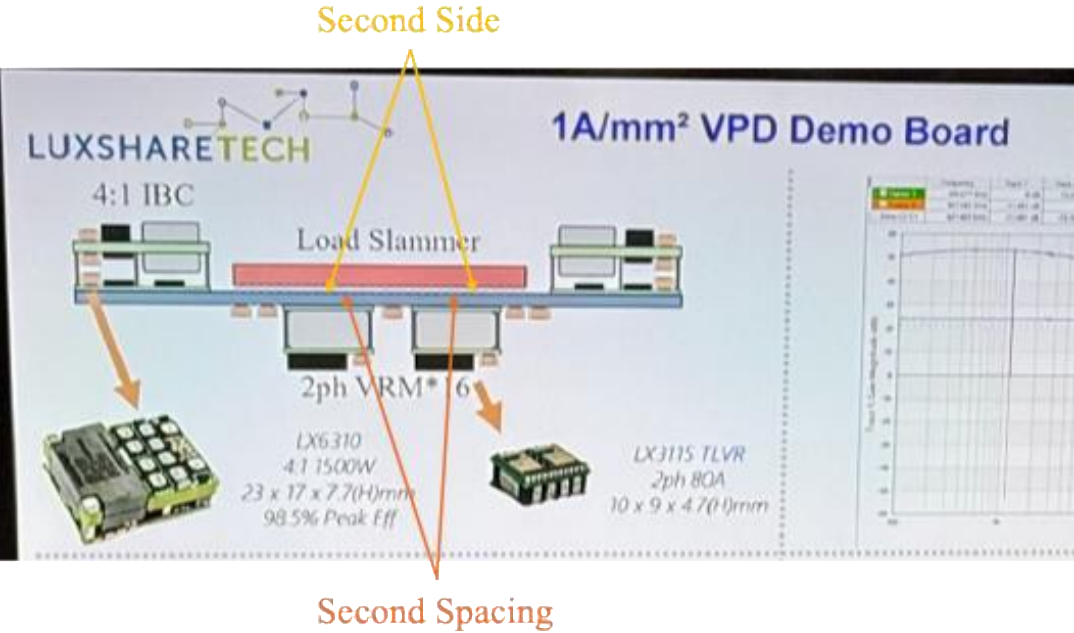
#	Limitation	Practice Evidence
		Ex. 32 at 2.
1.a	providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells,	The Luxshare VPD Module practices a method comprising “providing a power conversion module having conversion circuitry [red] arranged in a multi-cell structure having a plurality of conversion cells [blue].” 

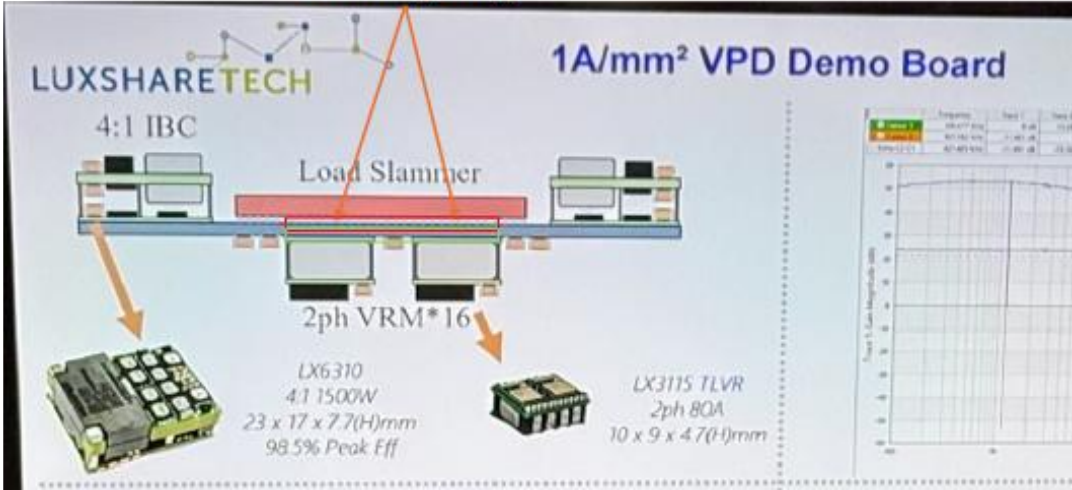
#	Limitation	Practice Evidence
		Ex. 32 at 4.
1.b	each conversion cell including one or more converter output terminals,	The Luxshare VPD Module practices a method comprising “each conversion cell including one or more converter output terminals [purple].”  Output terminals Ex. Luxhare at 2. See also 1.a.
1.c	the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of	The Luxshare VPD Module practices a method comprising “the conversion circuitry being configured to convert power received at a first voltage [red] for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage [green].”

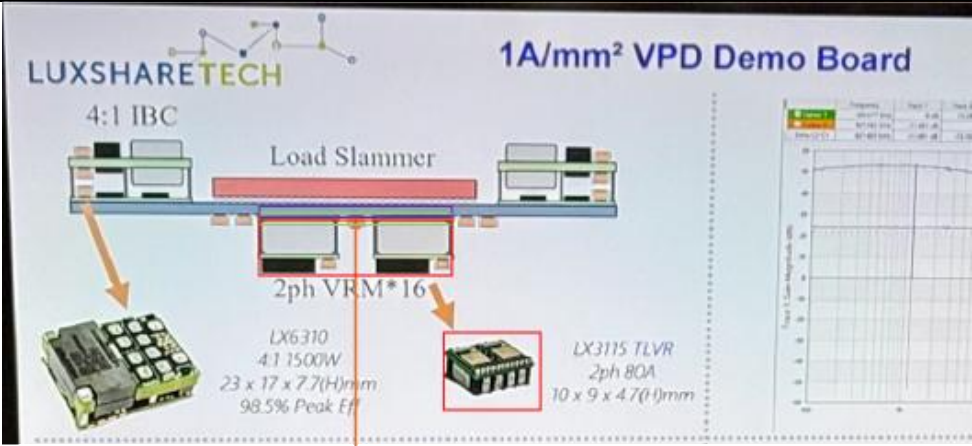
#	Limitation	Practice Evidence
	conversion cells at a second lower voltage,	The 4:1 IBC converts V_{in} from 54V to 13.5V. The conversion circuitry converts power received at 13.5V to the V_{out} at 0.8V.  The diagram illustrates the Luxshare VPD Module architecture. It features a 4:1 IBC (Input Block Converter) which converts the input voltage V_{in} from 54V to 13.5V. This is followed by a Load Slammer and a 2ph VRM*16 (2-phase Voltage Regulator Module). The module includes two main components: LX6310 (4:1 1500W, 23 x 17 x 7.7 (H)mm, 98.5% Peak Eff) and LX3115 TLVR (2ph 80A, 10 x 9 x 4.7 (H)mm). The End-to-End Efficiency Curve is shown for $V_{in}=54V$ and $V_{out}=0.8V$. Ex. Luxhare at 2.
1.d	the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals	The Luxshare VPD Module practices a method comprising “the converter output terminals being arranged in a first pattern [blue] having a first spacing [red] between the converter output terminals.”

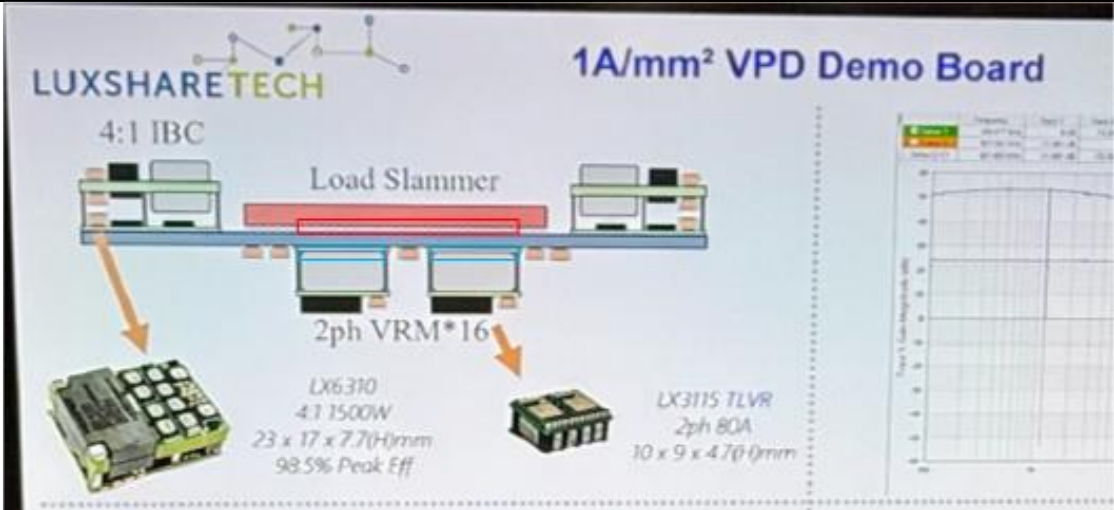
#	Limitation	Practice Evidence
		 Output terminals arranged in first pattern having a first spacing Ex. Luxhare at 2.
1.e	providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and	The Luxshare VPD Module practices a method comprising “providing an interconnection module [purple] having x-input terminals [green] arranged on a first side [orange] of the interconnection module in a pattern configured to align with the first pattern of converter output terminals.”

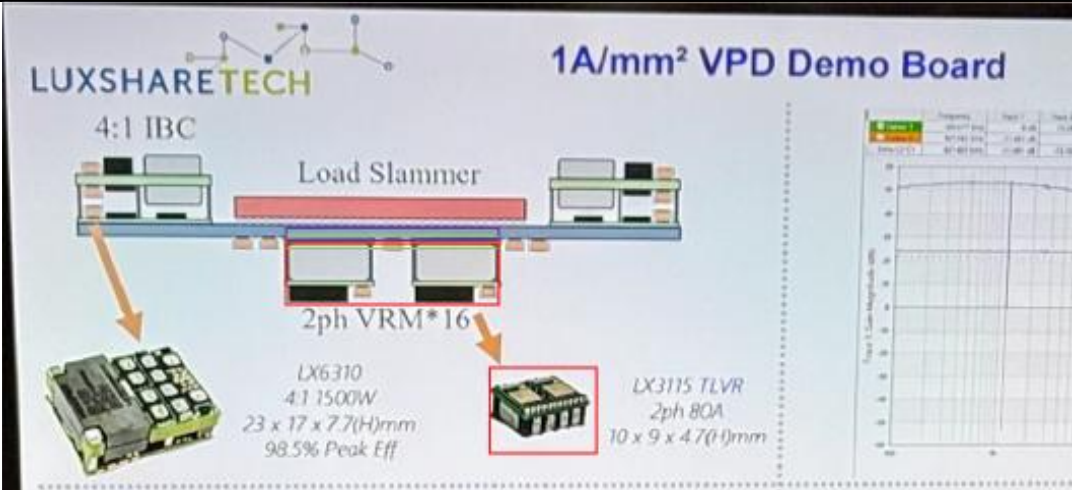
#	Limitation	Practice Evidence
		 Ex. Luxhare at 2.
1.f	x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals,	The Luxshare VPD Module practices a method comprising “x-output terminals arranged on a surface of a second opposite side [yellow] of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals [orange].” The x-output terminals of the interconnection module align with the power input terminals of the semiconductor device.

#	Limitation	Practice Evidence
		 Ex. Luxhare at 2.
1.g	the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;	The Luxshare VPD Module practices a method comprising “the second spacing [orange] being different than the first spacing, the x-input terminals [green] being electrically connected to the x-output terminals [red].”

#	Limitation	Practice Evidence
		 The diagram illustrates the assembly of a power conversion module to the first side of an interconnection module. Key components and labels include: LUXSHARETECH logo 4:1 IBC (Interconnection Block Connector) Load Slammer (red component) 2ph VRM*16 (Voltage Regulator Module) LX6310 4:1 1500W, 23 x 17 x 7.7(H)mm, 98.5% Peak Eff LX3115 TLVR 2ph 80A, 10 x 9 x 4.7(H)mm 1A/mm² VPD Demo Board title Second Spacing label with an orange triangle pointing to the Load Slammer - A graph on the right showing Power (W) vs Frequency (kHz). Ex. Luxhare at 2.
1.h	assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal;	The Luxshare VPD Module practices a method comprising “assembling the power conversion module [red] to the first side [orange] of the interconnection module [purple] with each converter output terminal mated with and electrically connected to a respective x-input terminal [green].”

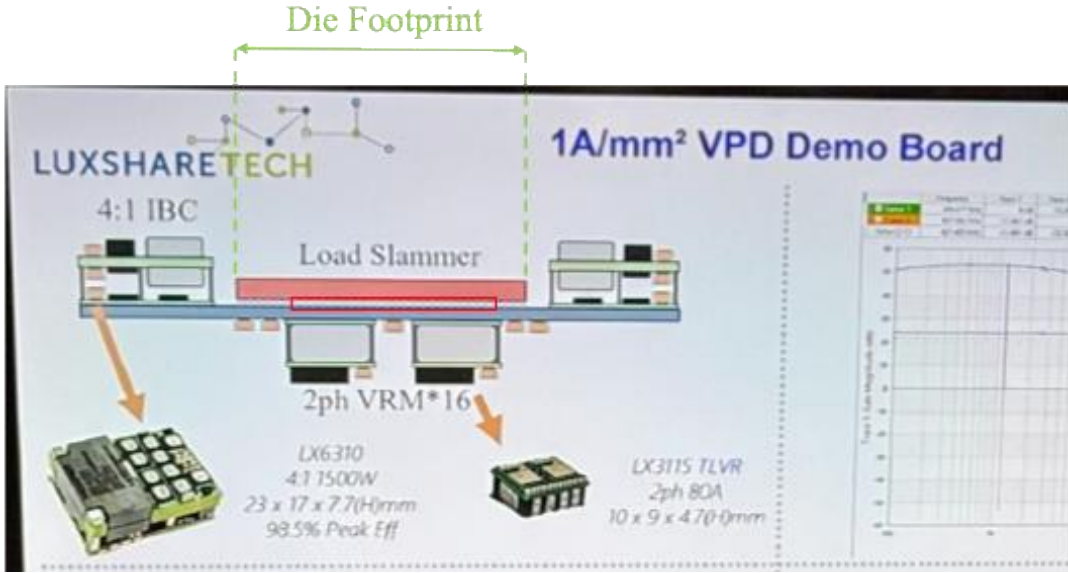
#	Limitation	Practice Evidence
		 First Side Ex. Luxhare at 2.
1.i	wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;	The Luxshare VPD Module practices a method comprising “wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern [blue] and the power input terminals of the semiconductor device arranged in the second pattern [red].”

#	Limitation	Practice Evidence
		 Ex. Luxhare at 2.
1.j	wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.	The Luxshare VPD Module practices a method comprising “wherein the power conversion module [red] and the interconnection module [purple] are stacked to form a module sandwich including the power conversion module [red] in a first layer, electrical connections between the converter output terminals and the x-input terminals [green] in a second layer, and the interconnection module [purple] in a third layer.”

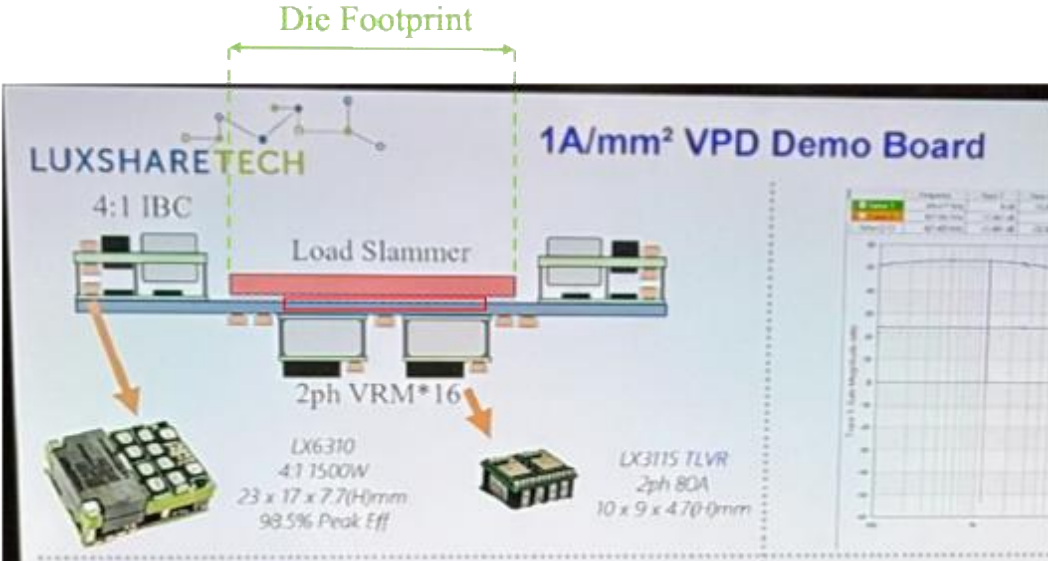
#	Limitation	Practice Evidence
		 Ex. Luxhare at 2.

Claim 2

#	Limitation	Practice Evidence
2.pre	A method of powering a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged	The Luxshare VPD Module practices a method comprising “[a] method of powering a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals [red] arranged on a surface of the die and having edges of the semiconductor die defining a die footprint [green], the semiconductor device having power input terminals arranged within the footprint of the die, the method comprising:”

#	Limitation	Practice Evidence
	within the footprint of the die, the method comprising:	 The diagram illustrates a 1A/mm² VPD Demo Board. It features a central 'Load Slammer' component connected to a '4:1 IBC' and '2ph VRM*16'. A 'Die Footprint' is indicated by a green dashed box. Below the board, two power modules are shown: an LX6310 4:1 1500W module (23 x 17 x 7.7(H)mm, 98.5% Peak Eff) and an LX3115 TLVR 2ph 80A module (10 x 9 x 4.7(H)mm). A graph on the right shows 'Power 1.0m Magnitude (mW)' vs. 'Frequency (kHz)'. Ex. Luxhare at 2. See also 1.pre.
2.a	providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells,	The Luxshare VPD Module practices a method comprising “providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells.” See 1.a.
2.b	each conversion cell including one or more converter output terminals,	The Luxshare VPD Module practices a method comprising “each conversion cell including one or more converter output terminals.” See 1.b.

#	Limitation	Practice Evidence
2.c	the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage,	The Luxshare VPD Module practices a method comprising “the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage.” <i>See 1.c.</i>
2.d	the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals;	The Luxshare VPD Module practices a method comprising “the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals.” <i>See 1.d.</i>
2.e	providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and	The Luxshare VPD Module practices a method comprising “providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals.” <i>See 1.e.</i>
2.f	x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern arranged within the die footprint and configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals,	The Luxshare VPD Module practices a method comprising “x-output terminals [red] arranged on a surface of a second opposite side of the interconnection module in a second pattern arranged within the die footprint [green] and configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals.” <i>See 1.f.</i> The Luxshare VPD Module practices a method comprising “x-output terminals [red] . . . arranged within the die footprint [green].”

#	Limitation	Practice Evidence
		 Ex. Luxhare at 2.
2.g	the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;	The Luxshare VPD Module practices a method comprising “the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals.” See 1.g.
2.h	assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated	The Luxshare VPD Module practices a method comprising “assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal.”

#	Limitation	Practice Evidence
	with and electrically connected to a respective x-input terminal;	<i>See 1.h.</i>
2.i	wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;	The Luxshare VPD Module practices a method comprising “wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern.” <i>See 1.i.</i>
2.j	wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.	The Luxshare VPD Module practices a method comprising “wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.” <i>See 1.j.</i>

Claim 40

#	Limitation	Practice Evidence
40.pre	An apparatus comprising:	Luxshare VPD Module is part of “[a]n apparatus.” <i>See 1.pre.</i>
40.a	a semiconductor device having power input terminals;	Luxshare VPD Module is part of an apparatus comprising “a semiconductor device having power input terminals.” <i>See 2.pre.</i>
40.b	a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells,	Luxshare VPD Module is part of an apparatus comprising “a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells.” <i>See 1.a.</i>
40.c	each conversion cell including one or more converter output terminals,	Luxshare VPD Module is part of an apparatus comprising “each conversion cell including one or more converter output terminals.” <i>See 1.b.</i>
40.d	the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage,	Luxshare VPD Module is part of an apparatus comprising “the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage.” <i>See 1.c.</i>
40.e	the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals; and	Luxshare VPD Module is part of an apparatus comprising “the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals.” <i>See 1.d.</i>

#	Limitation	Practice Evidence
40.f	an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and	Luxshare VPD Module is part of an apparatus comprising “an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals.” <i>See 1.e.</i>
40.g	x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals,	Luxshare VPD Module is part of an apparatus comprising “x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals.” <i>See 1.f.</i>
40.h	the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;	Luxshare VPD Module is part of an apparatus comprising “the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals.” <i>See 1.g.</i>
40.i	the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module;	Luxshare VPD Module is part of an apparatus comprising “the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module.” <i>See 1.h.</i>
40.j	wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input	Luxshare VPD Module is part of an apparatus “wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern.” <i>See 1.i.</i>

#	Limitation	Practice Evidence
	terminals of the semiconductor device arranged in the second pattern;	
40.k	wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.	Luxshare VPD Module is part of an apparatus “wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.” <i>See 1.j.</i>